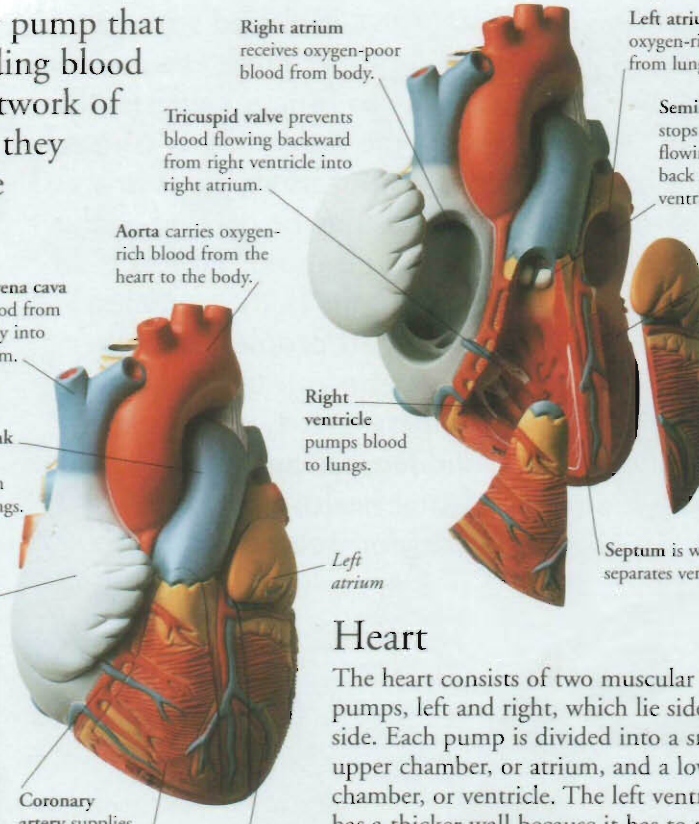
HEART AND CIRCULATORY SYSTEM

THE HEART IS A FIST-SIZED muscular pump that beats non-stop, 24 hours a day, sending blood around the body along a massive network of tubes called blood vessels. Together, they make up the circulatory system. The larger blood vessels divide repeatedly to form smaller vessels, which travel to every cell in the body, supplying them with oxygen from the lungs and nutrients from digested food, and carrying away waste. Blood helps defend the body against infection and also distributes heat around the body, helping to maintain its temperature.

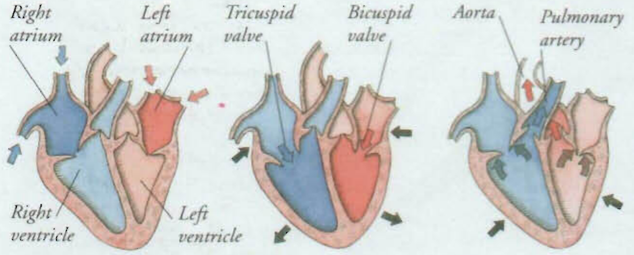


Heart

The heart consists of two muscular pumps, left and right, which lie side by side. Each pump is divided into a smaller upper chamber, or atrium, and a lower chamber, or ventricle. The left ventricle has a thicker wall because it has to pump blood around the body; the thinner-walled right ventricle pumps blood to the lungs.

How the heart beats

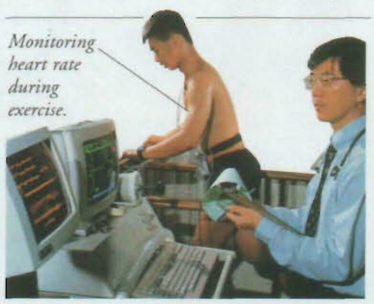
The wall of the heart is made of cardiac muscle that contracts automatically. The two halves of the heart beat together to pump blood around the body. Inside the heart, blood passes from the atria (upper chambers) to the ventricles (lower chambers). Valves ensure that blood cannot flow backwards through the heart. Each heartbeat is not a single contraction, but consists of three stages.



1 During the first stage (diastole) both the atria and the ventricles are relaxed. Blood flows into and fills both atria. The semilunar valves at the exit points of the ventricles are closed.

2 During the second stage (atrial systole) the tricuspid and bicuspid valves between the atria and the ventricles open. Both atria contract and squeeze blood into the ventricles below them.

3 During the third stage (ventricular systole) the ventricles contract to push blood out of the heart. The tricuspid and bicuspid valves close, while the semilunar valves open.

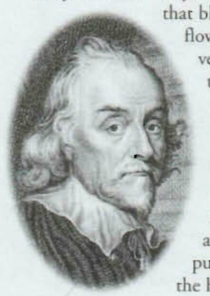


Heart rate

The heart normally beats about 70 times per minute. This is your heart rate. It changes according to the oxygen demands of the body. If you exercise, heart rate increases to pump more oxygen-carrying blood to your muscles.

William Harvey

English doctor William Harvey (1578–1657) was the first person to show that blood circulated around the body. Before Harvey, it was thought that blood ebbed and flowed along blood vessels rather like the tide coming in and going out. Harvey concluded that blood travelled in one direction only, and that it was pumped by the heart.



Blood

Blood is a liquid transport system that travels to every cell in the body. It supplies body cells with oxygen and nutrients, and carries away waste products. Blood consists of billions of blood cells floating in a yellowish liquid called plasma. There are three types of blood cells: red blood cells, white blood cells, and platelets. Red blood cells make up 99 per cent of all blood cells. A soft tissue inside bones called red marrow produces blood cells.

Platelets

Platelets are cell fragments that help stop blood leaking from injured blood vessels. If a blood vessel is damaged, platelets gather at the wound and stick to each other to form a plug.

White blood cells

White blood cells defend the body against infection. There are three main types. Granulocytes and monocytes engulf invading germs; lymphocytes release chemicals that destroy germs.

Red blood cells

Red blood cells are packed with a red substance called haemoglobin. Haemoglobin picks up oxygen in the lungs and releases it as blood passes through other parts of the body.